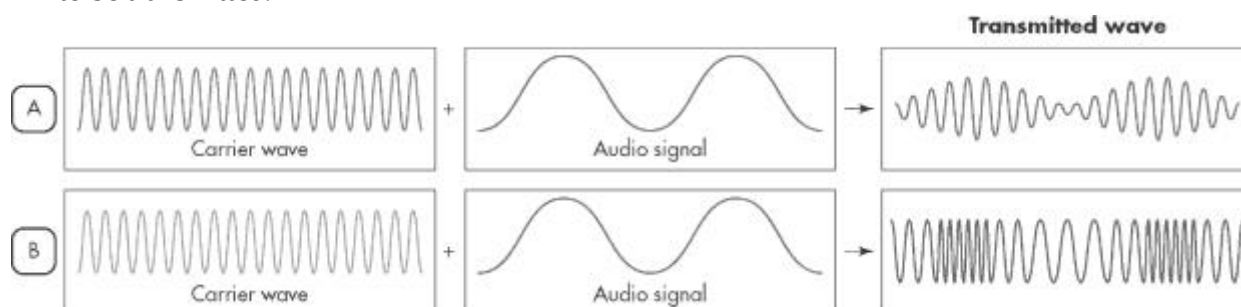


Communications

Answers

Transmitting radio waves

1. The following diagrams show how audio radio waves are combined with carrier waves in order to be transmitted.

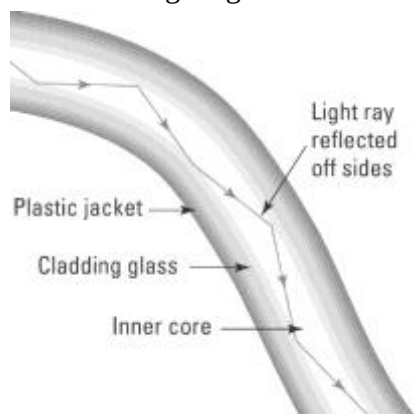


Identify which method corresponds to FM and which corresponds to AM transmission.

FM:B..... AM:A.....

Optical fibres

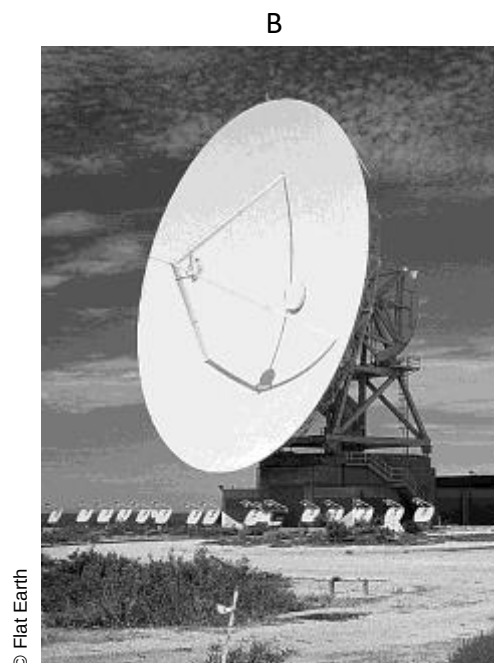
2. The following diagram shows a light ray moving through a section of optical fibre.



- (a) Name the process that allows the ray of light to be transmitted through the fibre.
Total internal reflection
- (b) Explain the purpose of the 'cladding glass'.
The cladding is denser than the glass of the inner core. The light rays reflect off this boundary rather than undergoing refraction. Thus no light energy is lost.
- (c) Explain how optical fibres are used in communication networks.
They can be laid underground or underwater and are used to carry long distance radio, TV or phone communications. More messages can be transmitted at one time than older technologies such as microwave or coaxial cable. Radio noise interference does not affect optical fibres.

Communication networks

3. Consider the following photographs.



- (a) Photograph A shows a microwave repeater station. Explain the purpose of these structures in our communication networks.

The antennas receive the microwave communication signals (TV, radio, phone and data) and then retransmit them on to the next repeater station. This prevents the signal fading away.

- (b) Photograph B shows a large dish antenna. Explain the purpose of this structure in our communication networks.

The dish antennas receive re-transmitted signals (radio, TV and phone) from geostationary satellites. Information can be communicated this way across the Australian continent.